PROGRAM

```
import numpy as np
from collections import Counter
# Dataset
X = np.array([
    [2, 60],
    [3, 65],
    [4, 70],
    [5, 75],
    [6, 80],
    [7, 85]
])
y = np.array([
    "Fail",
    "Fail",
    "Pass",
    "Pass",
    "Pass",
    "Pass"
])
# New student
new_student = np.array([4.5, 72])
# Calculate distances
distances = np.sqrt(np.sum((X - new_student) ** 2, axis=1))
print("Distances:")
for i, d in enumerate(distances):
    print("S" + str(i + 1), round(d, 3), y[i])
def knn_predict(X, y, test_point, k):
    distances = np.sqrt(
        np.sum((X - test_point) ** 2, axis=1)
    )
    nearest = np.argsort(distances)[:k]
    labels = y[nearest]
    prediction = Counter(labels).most_common(1)[0][0]
    return prediction
for k in [1, 3, 5]:
    prediction = knn_predict(
        X, y, new_student, k
    )
    print("Prediction for k =", k, ":", prediction)
```

OUTPUT

Distances:
S1 12.258 Fail
S2 7.159 Fail
S3 2.062 Pass
S4 3.041 Pass
S5 8.139 Pass
S6 13.238 Pass

Prediction for k = 1 : Pass
Prediction for k = 3 : Pass
Prediction for k = 5 : Pass